

```
# Instalação de pacotes necessários
pkgs <- c(
  "tidymodels",
  'glmnet',
  "rstanarm",
  "DescTools"
)

instalar <- pkgs[!pkgs %in% installed.packages()[, "Package"]]

if (length(instalar) > 0) {
  install.packages(instalar)
}
```

```
set.seed(42)
```

```
library(tidyverse)
library("tidymodels")
library('janitor')
library('ggcorrplot')
library(workflowsets)
library(rstanarm)
library(dplyr)
library(glmnet)
library(DescTools)
library(dplyr)
library(purrr)
library(yardstick)
```

```
Attaching core tidyverse packages          tidyverse 2.0.0
dplyr      1.2.1      readr      2.2.0
forcats    1.0.1      stringr   1.6.0
ggplot2    4.0.3      tibble    3.3.1
lubridate  1.9.5      tidyr     1.3.2
purrr      1.2.1
```

```
Conflicts              tidyverse_conflicts()
dplyr::filter() masks stats::filter()
dplyr::lag()      masks stats::lag()
```

Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```
Attaching packages          tidymodels 1.5.0
```

broom	1.0.12	rsample	1.3.2
dials	1.4.3	tailor	0.1.0
infer	1.1.0	tune	2.1.0
modeldata	1.5.1	workflows	1.3.0
parsnip	1.5.0	workflowsets	1.1.1
recipes	1.3.2	yardstick	1.4.0

```

Conflicts: tidymodels_conflicts()
scales::discard() masks purrr::discard()
dplyr::filter() masks stats::filter()
recipes::fixed() masks stringr::fixed()
dplyr::lag() masks stats::lag()
yardstick::spec() masks readr::spec()
recipes::step() masks stats::step()

```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

```
chisq.test, fisher.test
```

Loading required package: Rcpp

Attaching package: 'Rcpp'

The following object is masked from 'package:rsample':

```
populate
```

This is rstanarm version 2.32.2

- See <https://mc-stan.org/rstanarm/articles/priors> for changes to default priors!
- Default priors may change, so it's safest to specify priors, even if equivalent to the default
- For execution on a local, multicore CPU with excess RAM we recommend calling

```
options(mc.cores = parallel::detectCores())
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

```
expand, pack, unpack
```

Loaded glmnet 5.0

```
df <- read_csv("../data/titanic/train.csv") |>
  clean_names()
```

Rows: 891 Columns: 12

Column specification

Delimiter: ","

chr (5): Name, Sex, Ticket, Cabin, Embarked

dbl (7): PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare

Use `spec()` to retrieve the full column specification for this data.

Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
#preprocessamento
df <- janitor::clean_names(df)
```

```
df_num <- df |>
  dplyr::select(where(is.numeric))
```

```
cor_matrix <- cor(
  df_num,
  use = "complete.obs",
  method = "pearson"
)
```

```
ggcorrplot(
```

```

cor_matrix,
type = "lower",
lab = TRUE,
lab_size = 4,
colors = c("#b2182b", "white", "#2166ac"),
outline.col = "gray70"
) +
theme_minimal(base_size = 14) +
theme(
  axis.text.x = element_text(
    angle = 90,
    hjust = 1,
    vjust = 0.5
  ),
  axis.text.y = element_text(size = 8)
)

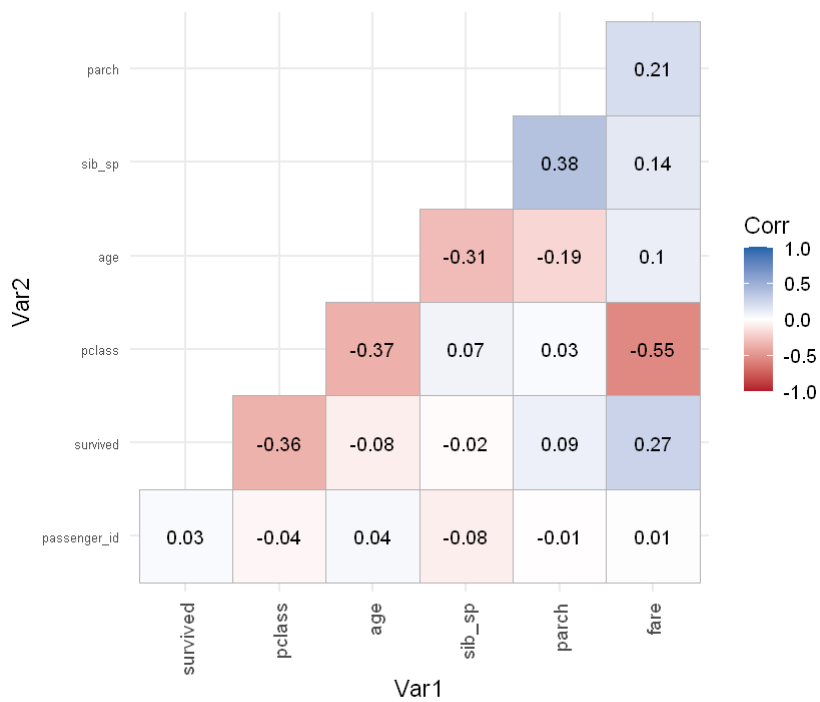
```

Warning message:

```

"`aes_string()` was deprecated in ggplot2 3.0.0.
Please use tidy evaluation idioms with `aes()`.
See also `vignette("ggplot2-in-packages")` for more information.
The deprecated feature was likely used in the ggcorrplot package.
Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>."

```



```
cat("Idades faltantes:", sum(is.na(df$age)), "\n")
idade_mediana <- median(df$age, na.rm = TRUE)

df <- df %>%
  mutate(age = ifelse(is.na(age), idade_mediana, age))

idade_mediana
```

Idades faltantes: 177

28

```
cat("Embarque faltante:", sum(is.na(df$embarked)), "\n")

puerto_moda <- Mode(df$embarked, na.rm = TRUE)

df$embarked[is.na(df$embarked)] <- puerto_moda
```

Embarque faltante: 2

```
df$HasCabin <- as.integer(!is.na(df$cabin))
```

```
df <- select(df, -cabin)
```

```
df <- df %>%  
  mutate(fare = ifelse(is.na(fare), median(fare, na.rm = TRUE), fare))
```

```
df <- df %>%  
  mutate(  
    sex = ifelse(sex == "male", 0, 1),  
    embarked = case_when(  
      embarked == "S" ~ 0,  
      embarked == "C" ~ 1,  
      embarked == "Q" ~ 2  
    )  
  )
```

```
df <- df %>%  
  select(-name, -ticket, -passenger_id)
```

```
df_num <- df |>  
  dplyr::select(where(is.numeric))
```

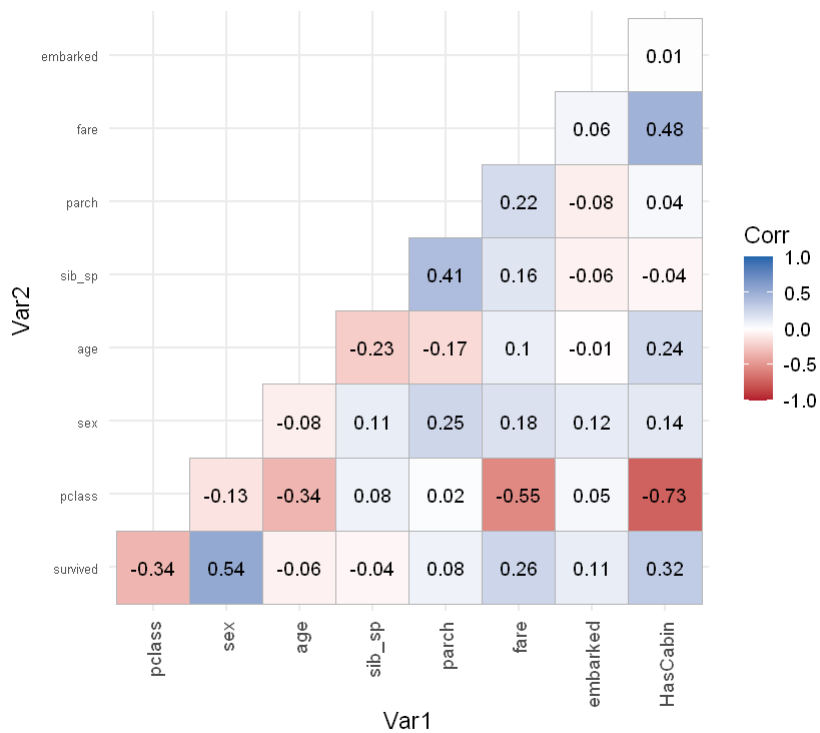
```
cor_matrix <- cor(  
  df_num,  
  use = "complete.obs",  
  method = "pearson"  
)
```

```
ggcorrplot(  
  cor_matrix,  
  type = "lower",  
  lab = TRUE,  
  lab_size = 4,  
  colors = c("#b2182b", "white", "#2166ac"),  
  outline.col = "gray70"  
) +
```

```

theme_minimal(base_size = 14) +
theme(
  axis.text.x = element_text(
    angle = 90,
    hjust = 1,
    vjust = 0.5
  ),
  axis.text.y = element_text(size = 8)
)

```



```

split <- initial_split(
  df,
  prop = 0.75,
  strata = survived
)

train_data <- training(split)
test_data <- testing(split)

```

```
train_data$survived <- as.factor(train_data$survived)
```

```
train_data$survived <- as.factor(train_data$survived)
test_data$survived <- as.factor(test_data$survived)
train_data$survived <- factor(train_data$survived, levels = c("0", "1"))
test_data$survived <- factor(test_data$survived, levels = c("0", "1"))
```

```
log_spec <- logistic_reg() |>
  set_engine("glm") |>
  set_mode("classification")
```

```
rec_all <- recipe(survived ~ ., data = train_data)

rec_no_sex <- recipe(survived ~ ., data = train_data) |>
  step_rm(sex)

rec_sex_only <- recipe(survived ~ sex, data = train_data)
```

```
wf_logit <- workflow_set(
  preproc = list(
    all_features = rec_all,
    no_sex       = rec_no_sex,
    sex_only     = rec_sex_only
  ),
  models = list(
    logit = log_spec
  )
)
```

```
wf_fitted <- wf_logit %>%
  mutate(
    fit = purrr::map(
      info,
      ~ fit(.x$workflow[[1]], data = train_data)
    )
  )
```

```
pred_list <- wf_fitted$fit %>%
  map(~ bind_cols(
    predict(.x, test_data, type = "class"),
```

```

  predict(.x, test_data, type = "prob"),
  test_data
))

```

```

pred_all <- wf_fitted$wflow_id %>%
  map2(pred_list, ~ mutate(.y, model = .x)) %>%
  bind_rows()

```

```

metrics_class <- pred_all %>%
  group_by(model) %>%
  metric_set(
    accuracy,
    precision,
    recall
  )(
    truth = survived,
    estimate = .pred_class
  )

```

```

metrics_auc <- pred_all %>%
  group_by(model) %>%
  roc_auc(
    truth = survived,
    .pred_1,
    event_level = "second"
  )

```

```

metrics_final <- bind_rows(metrics_class, metrics_auc)
metrics_final

```

A tibble: 12 × 4

model <chr>	.metric <chr>	.estimator <chr>	.estimate <dbl>
all_features_logit	accuracy	binary	0.8214286
no_sex_logit	accuracy	binary	0.7455357
sex_only_logit	accuracy	binary	0.7991071
all_features_logit	precision	binary	0.8310811
no_sex_logit	precision	binary	0.7396450
sex_only_logit	precision	binary	0.8039216
all_features_logit	recall	binary	0.8913043

model <chr>	.metric <chr>	.estimator <chr>	.estimate <dbl>
no_sex_logit	recall	binary	0.9057971
sex_only_logit	recall	binary	0.8913043
all_features_logit	roc_auc	binary	0.8643411
no_sex_logit	roc_auc	binary	0.7554348
sex_only_logit	roc_auc	binary	0.7712336

```
model_summary <- tibble(
  model = wf_fitted$wflow_id,
  summary = map(
    wf_fitted$fit,
    ~ summary(extract_fit_parsnip(.x)$fit)
  )
)
for (i in seq_len(nrow(model_summary))) {
  cat("\n=====\n")
  cat("Modelo:", model_summary$model[i], "\n")
  cat("=====\n\n")
  print(model_summary$summary[[i]])
}
```

```
=====
Modelo: all_features_logit
=====
```

```
Call:
stats::glm(formula = ..y ~ ., family = stats::binomial, data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.6416012	0.5994780	2.738	0.00617	**
pclass	-0.9388130	0.1883676	-4.984	6.23e-07	***
sex	2.7062567	0.2303986	11.746	< 2e-16	***
age	-0.0373637	0.0089442	-4.177	2.95e-05	***
sib_sp	-0.3370826	0.1254324	-2.687	0.00720	**
parch	-0.1517747	0.1331173	-1.140	0.25422	
fare	0.0005517	0.0026691	0.207	0.83626	
embarked	0.2337466	0.1639045	1.426	0.15384	
HasCabin	0.5176190	0.3360462	1.540	0.12348	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 888.31 on 666 degrees of freedom
Residual deviance: 592.93 on 658 degrees of freedom
AIC: 610.93

Number of Fisher Scoring iterations: 5

=====
Modelo: no_sex_logit
=====

Call:

stats::glm(formula = ..y ~ ., family = stats::binomial, data = data)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.155636	0.538406	4.004	6.23e-05	***
pclass	-0.822474	0.165761	-4.962	6.98e-07	***
age	-0.037666	0.007774	-4.845	1.26e-06	***
sib_sp	-0.220017	0.101914	-2.159	0.03086	*
parch	0.223885	0.117041	1.913	0.05576	.
fare	0.001821	0.002494	0.730	0.46544	
embarked	0.420042	0.135461	3.101	0.00193	**
HasCabin	0.585459	0.292334	2.003	0.04521	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 888.31 on 666 degrees of freedom
Residual deviance: 765.48 on 659 degrees of freedom
AIC: 781.48

Number of Fisher Scoring iterations: 4

=====

Modelo: sex_only_logit

=====

Call:

stats::glm(formula = ..y ~ ., family = stats::binomial, data = data)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.4741	0.1247	-11.82	<2e-16 ***
sex	2.4606	0.1907	12.90	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 888.31 on 666 degrees of freedom
Residual deviance: 691.99 on 665 degrees of freedom
AIC: 695.99

Number of Fisher Scoring iterations: 4